

PLANT ANATOMY

Comprehensive Revision Notes aligned with NEB Board Guidelines & Common Entrance Examination (CEE) High-Yield Blueprint

1.1 Meristematic Tissues

Meristematic tissues consist of actively dividing, immature, and undifferentiated cells that retain the continuous capacity for cell division. These cells possess thin cellulosic primary walls, dense cytoplasm, prominent large central nuclei, and generally lack vacuoles or intercellular spaces.

Classification based on Position

- **Apical Meristem:** Located at the growing apex of roots and shoots (Root Apical Meristem - RAM, and Shoot Apical Meristem - SAM). It is primarily responsible for the elongation of the plant body, known as *primary growth*.
- **Intercalary Meristem:** Derived from the apical meristem but separated from it by patches of permanent tissues. Typically found at the base of nodes, internodes, or leaf bases (e.g., in grasses and mints). It helps in regenerating components removed or grazed by herbivores.
- **Lateral Meristem:** Arranged parallel to the lateral axis of the plant organs (e.g., Vascular Cambium and Cork Cambium/Phellogen). It is responsible for increasing the girth or thickness of the plant axis via *secondary growth*.

 **CEE Focus Point:** Both apical and intercalary meristems are *primary meristems* because they appear early in a plant's life and contribute to the primary plant body. Most lateral meristems are secondary in origin (formed later from permanent tissues).

1.2 Permanent Tissues

Permanent tissues are composed of mature cells that have completely or partially lost their capacity to divide, having attained a definite shape, size, configuration, and specialized functional role through differentiation.

A. Simple Permanent Tissues (Homogeneous Tissues)

Formed of structurally and functionally identical cell types. There are three main types:

- **Parenchyma:** Universal, living, thin-walled isodiametric cells with intercellular spaces. Cell walls are purely cellulosic. Function: Storage, assimilation. Specialized forms include *Chlorenchyma* (photosynthetic parenchyma) and *Aerenchyma* (buoyancy providing parenchyma with large air cavities, found in hydrophytes).

- **Collenchyma:** Living mechanical tissue. Cells feature characteristic uneven wall thickenings at the corners composed of *pectin, cellulose, and hemicellulose*. It provides tensile strength with flexibility to young growing regions.
- **Sclerenchyma:** Dead mechanical tissue at maturity, characterized by highly thickened, heavily lignified cell walls with extremely narrow lumina. It is completely devoid of protoplasts. Divided into elongated *Fibres* and hard, irregular *Sclereids* (stone cells/grit cells responsible for the gritty texture in guava, pear, and sapota fruits).

⚡ **CEE Focus Point:** *Collenchyma is strictly absent in roots and monocot plants.* It occurs selectively in the hypodermis of dicot stems.

B. Complex Permanent Tissues (Heterogeneous Tissues)

Composed of multiple types of cells that function as a single coordinated structural unit to perform conduction.

1. Xylem (Water and Mineral Conducting Tissue)

- **Tracheids:** Elongated, dead, tube-like cells with tapering ends and lignified walls. Present in all vascular plants.
- **Vessels (Tracheae):** Long, cylindrical, vessel-like syncytial structures with perforated end-walls. *Vessels are a diagnostic feature of Angiosperms*; they are completely absent in Gymnosperms and Pteridophytes (with rare exceptions like *Gnetum*).
- **Xylem Fibres:** Highly thickened dead structures providing structural mechanical support.
- **Xylem Parenchyma:** The **only living component** of xylem; primarily involved in the short-distance lateral conduction of water and storage of starch/fats.

2. Phloem (Food Conducting Tissue / Bast)

- **Sieve Tubes:** Long, tube-like structures arranged longitudinally, associated with sieve plates. Crucially, mature sieve tube elements *possess a living protoplast but lack a nucleus*.
- **Companion Cells:** Specialized parenchymatous cells structurally and functionally tied to sieve tubes via plasmodesmata. Their nuclei regulate the pressure gradients within the sieve tubes.
- **Phloem Parenchyma:** Absent in most monocotyledons. Stores food material.
- **Phloem Fibres (Bast Fibres):** Sclerenchymatous fibers that represent the **only dead component** of primary phloem. Extensively utilized commercially (e.g., jute, flax, hemp).

C. Secretory Tissues

- **Laticiferous Tissues:** Specialized tube-like structures containing latex (e.g., *Ficus elastica*, *Euphorbia*, *Argemone*).
- **Glandular Tissues:** Specialized internal or external structures. Examples include the digestive glands of insectivorous plants (e.g., *Nepenthes*), oil glands in *Citrus*, and resin ducts in *Pinus*.

1.3 Anatomy of Root, Stem, and Leaf

To differentiate systematically between the primary inner structures of plant organs, the key diagnostic anatomical criteria used in NEB and CEE structural profiles are detailed below:

Anatomical Feature	Dicot Root	Monocot Root	Dicot Stem	Monocot Stem
Vascular Bundle Type	Radial arrangement	Radial arrangement	Conjoint, Collateral, Open	Conjoint, Collateral, Closed
Xylem Orientation	Exarch (Protoxylem outside)	Exarch (Protoxylem outside)	Endarch (Protoxylem inside)	Endarch (Protoxylem inside)
Number of Bundles	Diarch to Tetrarch (2 to 4)	Polyarch (Greater than 6)	Fixed number, in a single ring	Numerous, Scattered
Hypodermis	Absent	Absent	Collenchymatous	Sclerenchymatous
Pith (Medulla)	Absent or poorly developed	Large and well-developed	Well-developed, distinct	Absent (Undifferentiated)
Bundle Sheath	Absent	Absent	Absent	Sclerenchymatous sheath present
Special Characters	Casparian strips in endodermis	Pronounced passage cells	Ring arrangement of bundles	'Y' or 'V' shaped xylem; water cavity

⚡ High-Yield Key Terms Defined:

- **Radial:** Xylem and phloem sit on completely alternating radii.
- **Conjoint, Collateral:** Xylem and phloem sit on the same radius, with phloem located towards the outside.
- **Open Bundle:** Cambium strips exist between the xylem and phloem (capable of secondary growth).
- **Closed Bundle:** Intrafascicular cambium is completely absent (incapable of normal secondary growth).

1.3.5 & 1.3.6 Internal Structure of Leaves

- **Dorsiventral Leaf (Dicot Leaf):** Typically oriented horizontally. Stomata are located predominantly on the lower epidermis (**Hypostomatic**). Mesophyll tissue is clearly differentiated into upper elongated, tightly packed **palisade parenchyma** and lower loosely arranged, vascularized **spongy parenchyma**.
- **Isobilateral Leaf (Monocot Leaf):** Typically oriented vertically. Stomata are distributed equally on both the upper and lower surfaces (**Amphistomatic**). Mesophyll tissue remains completely uniform and undifferentiated.

⚡ **CEE Focus Point:** Monocot leaf borders display large, specialized, empty, colorless epidermal cells called **Bulliform cells (or Motor cells)**. Under water stress, they lose turgidity, causing leaves to roll inward to drastically reduce transpiration rates.

1.4 Secondary Growth in Dicot Stem

Secondary growth refers to the structural increase in the thickness or girth of a plant axis, driven by the activity of lateral meristems (Vascular Cambium and Cork Cambium). It is a defining feature of Dicotyledons and Gymnosperms.

1. Formation of the Vascular Cambium Ring

The primary vascular bundles in dicot stems contain a strip of meristem known as **Intrafascicular Cambium** (primary origin). During secondary growth, cells of the medullary rays adjacent to these strips differentiate into **Interfascicular Cambium** (secondary origin). These two types link up to form a continuous, internal **Vascular Cambium Ring**.

2. Meristematic Activity and Wood Differentiation

The vascular cambium ring divides periclinally, cutting off new cells on both sides. Cells cut off toward the interior differentiate into **Secondary Xylem** (which constitutes the bulk of the wood), while cells cut off toward the exterior differentiate into **Secondary Phloem**. The cambium is significantly more active on its inner side than its outer side.

3. Seasonal Activity: Spring Wood vs. Autumn Wood

- **Spring Wood (Early Wood):** Formed during spring when cambial activity is highly intense. It produces numerous xylem vessels with wide cavities and thinner walls. It features lower structural density and a lighter appearance.
- **Autumn Wood (Late Wood):** Formed during winter/autumn when cambial activity slows down. It produces fewer, narrower vessels with thick walls. It features high density and a darker appearance.

The concentric alternation of one spring wood band and one autumn wood band forms a distinct **Annual Ring (Growth Ring)**. Counting these rings allows for the estimation of a tree's age, a scientific discipline called **Dendrochronology**.

4. Heartwood vs. Sapwood

- **Heartwood (Duramen):** The central, older region of secondary xylem. It is non-functional for water transport due to the deposition of organic compounds (tannins, resins, gums, oils) and the formation of balloon-like blockages called **Tyloses** in vessel lumina. It is highly durable, dark in color, and provides mechanical rigidity.
- **Sapwood (Alburnum):** The peripheral, younger region of secondary xylem. It is light-colored, functional, and actively conducts water and mineral salts from roots to leaves.

5. Extra-stelar Secondary Growth (Bark Mechanics)

To accommodate the continuous internal expansion of secondary xylem, the outer cortical tissues break down. A secondary lateral meristem called the **Cork Cambium (Phellogen)** differentiates in the outer cortex.

- Phellogen cuts off heavily suberized protective cells called **Cork (Phellem)** on the outside.
- Phellogen cuts off parenchymatous cells called the **Secondary Cortex (Phelloderm)** on the inside.
- The term **Periderm** collectively refers to these three layers: *Phellem + Phellogen + Phelloderm*.
- **Bark:** A non-technical term encompassing all tissues exterior to the vascular cambium ring (includes Periderm, Cortex, Primary, and Secondary Phloem).
- **Lenticels:** Lens-shaped aerating openings formed in the periderm of woody stems when phellogen cuts off loosely arranged complementary cells instead of tightly packed cork. They facilitate essential gaseous exchange.